

Лабораторная работа №2

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Тема: Настройка протокола STP (IEEE 802.1D)

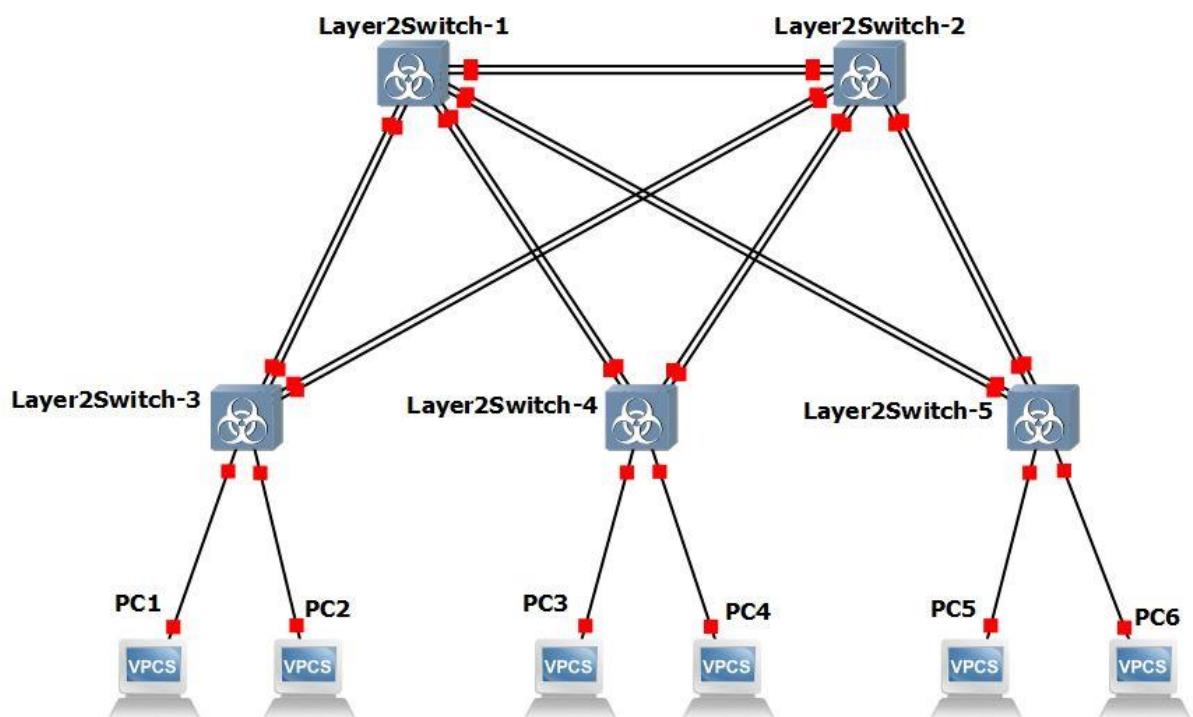
Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

Полезная информация: схема сохранена на сервере в проекте под именем Menzhulin-lab2-template, можно использовать кнопку Duplicate

1) Для заданной на схеме **schema-lab2** сети, состоящей из управляемых коммутаторов и персональных компьютеров

настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета



Для принудительного указания корневого коммутатора нужно изменить его BID (его приоритет), т.к. корневой коммутатор определяется по наименьшему приоритету. Приоритет по умолчанию равен 32768 + номер VLAN. Изначально у всех свитчей приоритет равен 32769

Изменим конфигурацию коммутатора 1.

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096

vIOS-L2-01#show spanning-tree

vIOS-L2-01#write memory

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#show spanning-tree
[
^
% Invalid input detected at '^' marker.

vIOS-L2-01(config)#^Z
vIOS-L2-01#
*Aug 2 02:47:03.912: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#show spanning-tree
[
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address    0c39.32cd.0000
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4097 (priority 4096 sys-id-ext 1)
             Address    0c39.32cd.0000
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Desg FWD 4        128.1   Shr
Gi0/1                    Desg FWD 4        128.2   Shr
Gi0/2                    Desg FWD 4        128.3   Shr
Gi0/3                    Desg FWD 4        128.4   Shr
Gi1/0                    Desg FWD 4        128.5   Shr
Gi1/1                    Desg FWD 4        128.6   Shr
Gi1/2                    Desg FWD 4        128.7   Shr
Gi1/3                    Desg LRN 4        128.8   Shr

vIOS-L2-01#rite memory
[
^
% Invalid input detected at '^' marker.

vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5276 bytes to 2025 bytes[OK]
*Aug 2 02:47:33.561: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Aug 2 02:47:34.443: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
vIOS-L2-01#
```

Корневым становится SW1

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

Ниже приведены результаты выполнения команды ping и полученные ответы от каждого компьютера. Видно, что все компьютеры доступны друг другу.

PC1

```
[PC1> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=0.812 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=0.607 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=2.608 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=5.343 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=1.343 ms

[PC1> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=17.678 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=6.365 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=7.803 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=8.342 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=3.096 ms

[PC1> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=5.234 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=5.499 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=5.226 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=6.906 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=10.683 ms

[PC1> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=20.173 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=21.949 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=9.502 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=7.046 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=9.588 ms

[PC1> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=15.409 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=4.777 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=4.981 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=21.423 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=8.457 ms
```

PC2

```
[PC2> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=9.547 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=3.694 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=2.487 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=3.813 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=7.524 ms

[PC2> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=6.523 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=10.847 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=7.117 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=3.840 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=12.529 ms

[PC2> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=7.785 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=8.308 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=10.467 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=6.704 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=8.031 ms

[PC2> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=7.284 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=6.231 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=4.916 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=12.702 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=24.576 ms

[PC2> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=12.889 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=3.025 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=4.500 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=5.658 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=13.831 ms
```

PC3

[PC3> ping 192.168.0.1

```
84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=19.558 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=8.849 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=8.320 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=6.350 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=14.398 ms
```

[PC3> ping 192.168.0.2

```
84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=8.887 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=27.463 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=11.005 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=6.778 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=16.247 ms
```

[PC3> ping 192.168.0.4

```
84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=2.166 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=6.760 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=4.455 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=1.524 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=4.554 ms
```

[PC3> ping 192.168.0.5

```
84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=15.191 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=6.179 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=7.085 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=4.750 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=15.660 ms
```

[PC3> ping 192.168.0.6

```
84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=15.206 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=4.301 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=17.859 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=8.054 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=13.958 ms
```

PC4

[PC4> ping 192.168.0.1

```
84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=22.600 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=6.242 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=16.212 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=7.181 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=8.354 ms
```

[PC4> ping 192.168.0.2

```
84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=15.013 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=16.171 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=8.391 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=9.005 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=6.195 ms
```

[PC4> ping 192.168.0.3

```
84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=2.673 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=4.238 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=0.845 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=4.625 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=5.629 ms
```

[PC4> ping 192.168.0.5

```
84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=14.629 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=8.380 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=6.254 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=14.608 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=4.696 ms
```

[PC4> ping 192.168.0.6

```
84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=16.178 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=5.401 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=9.300 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=9.496 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=9.448 ms
```

PC5

[PC5> ping 192.168.0.1

```
84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=9.083 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=7.871 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=12.015 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=5.012 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=8.692 ms
```

[PC5> ping 192.168.0.2

```
84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=19.751 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=10.262 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=12.305 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=6.431 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=8.130 ms
```

[PC5> ping 192.168.0.3

```
84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=15.394 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=7.527 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=5.447 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=6.069 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=8.107 ms
```

[PC5> ping 192.168.0.4

```
84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=3.557 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=15.173 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=12.300 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=9.688 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=9.665 ms
```

[PC5> ping 192.168.0.6

```
84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=3.019 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=1.487 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=8.617 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=2.030 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=1.801 ms
```

PC6

[PC6> ping 192.168.0.1

```
84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=8.986 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=5.676 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=10.320 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=10.741 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=8.017 ms
```

[PC6> ping 192.168.0.2

```
84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=7.060 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=6.323 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=4.439 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=8.882 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=11.750 ms
```

[PC6> ping 192.168.0.3

```
84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=3.829 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=15.324 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=12.004 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=13.291 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=15.650 ms
```

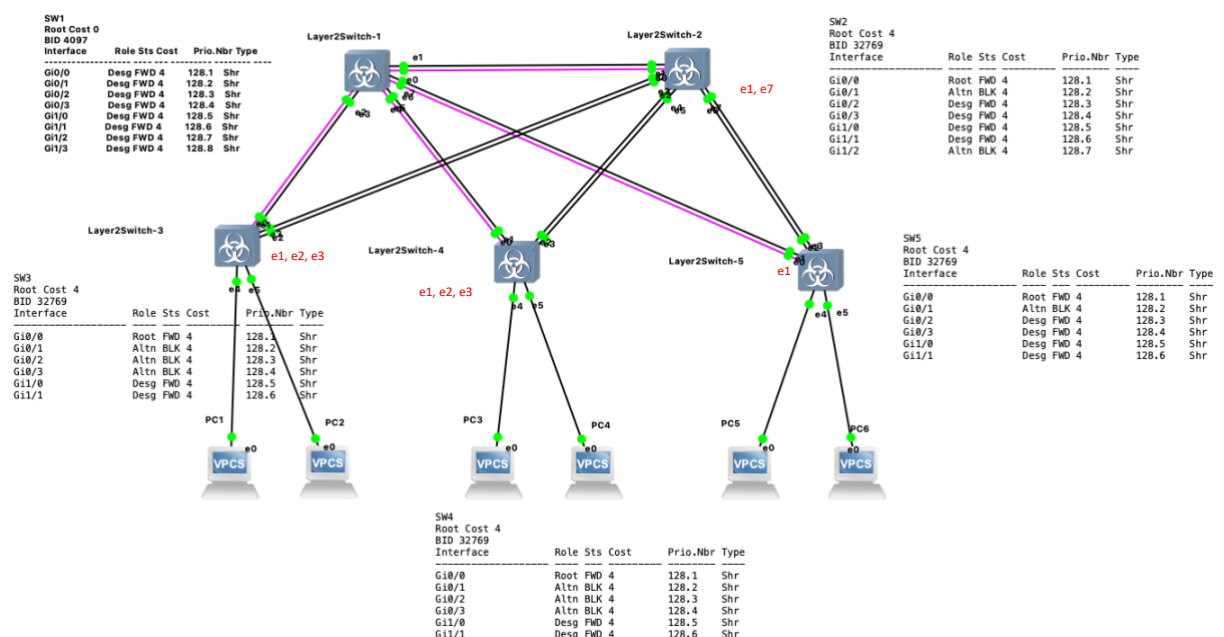
[PC6> ping 192.168.0.4

```
84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=4.376 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=12.644 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=8.528 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=9.838 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=6.663 ms
```

[PC6> ping 192.168.0.5

```
84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=6.915 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=3.066 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=2.650 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=1.271 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=4.827 ms
```

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

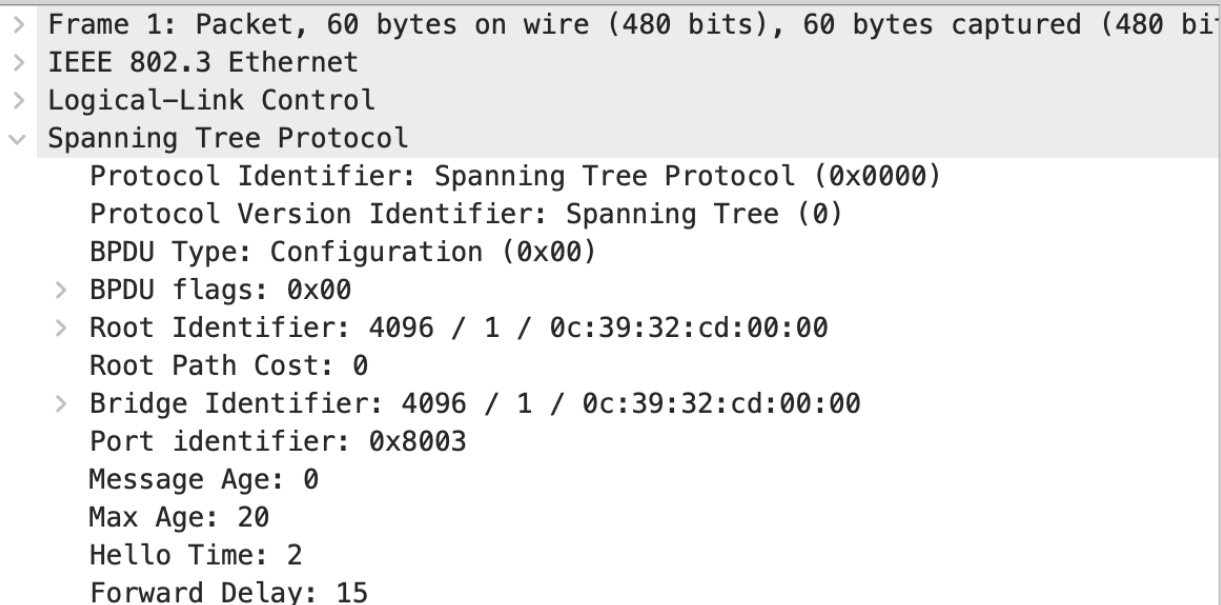


Розовым цветом указаны пути передачи данных.

Красным отмечены заблокированные порты.

Рядом со свитчами указаны корневая стоимость, BID, состояние и стоимость каждого порта.

4) При помощи Wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет



```
> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bi
> IEEE 802.3 Ethernet
> Logical-Link Control
✓ Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
    > BPDU flags: 0x00
    > Root Identifier: 4096 / 1 / 0c:39:32:cd:00:00
        Root Path Cost: 0
    > Bridge Identifier: 4096 / 1 / 0c:39:32:cd:00:00
        Port identifier: 0x8003
        Message Age: 0
        Max Age: 20
        Hello Time: 2
        Forward Delay: 15
```

Все пакеты, отправленные корневым свитчем (SW1), имеют стоимость 0.

Коммутаторы пересылающие эти сообщения дальше указывают корневую стоимость 4, меняют BID отправителя на свой BID.

Неназначенные порты только слушают BPDU.

Назначенные порты пересылают все пакеты дальше по сети.

В заголовке STP:

- идентификатор протокола
- версия протокола
- тип BPDU сообщения
- флаги
- BID корневого коммутатора
- Время жизни сообщения
- Максимальное время жизни сообщения
- Период между отправками hello пакетов
- Время задержки пересылки

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл


```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#interface GigabitEthernet0/0
```

```
vIOS-L2-01(config-if)#spanning-tree cost 7
```

```
vIOS-L2-01(config-if)#exit
```

```
[vIOS-L2-01#show spanning-tree
```

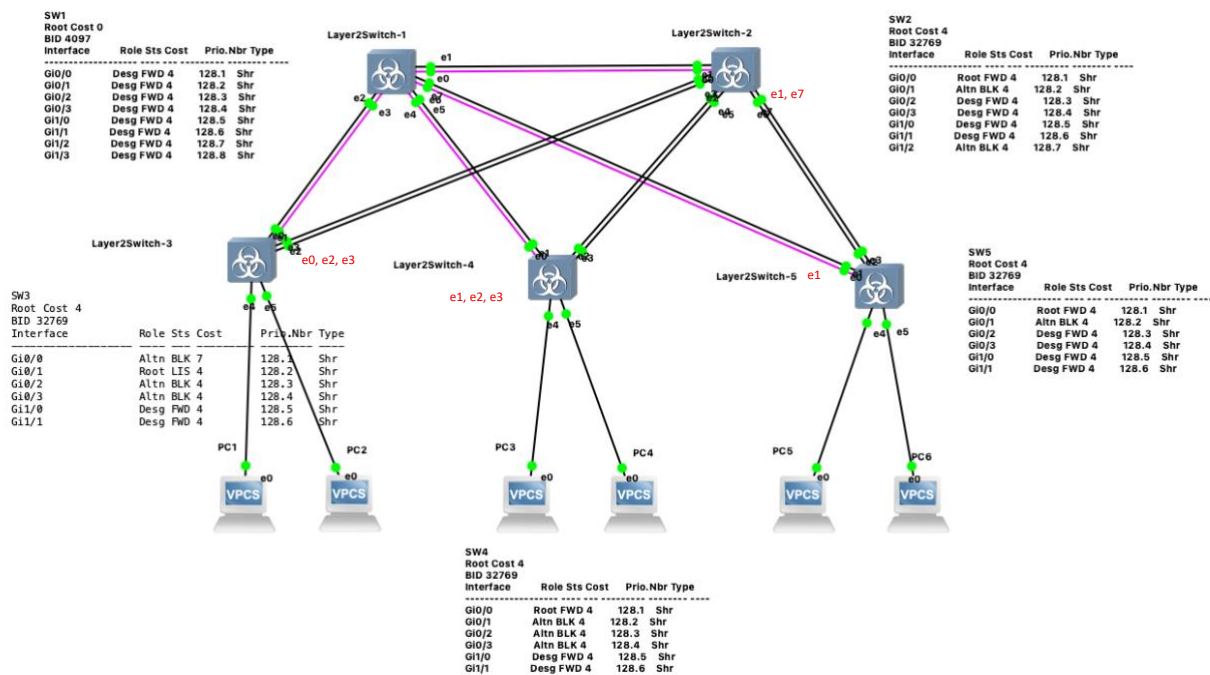
```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID      Priority    4097
             Address    0c39.32cd.0000
             Cost      4
             Port      2 (GigabitEthernet0/1)
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID    Priority    32769 (priority 32768 sys-id-ext 1)
             Address    0c4c.31fb.0000
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 15 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Altn	BLK	7	128.1	Shr
Gi0/1	Root	LIS	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr



Изменился линк передачи данных от корневого свитча до свитча 3, т.к. стоимость корня должна быть ниже.

6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

7*) Опциональное задание: заменить STP на RSTP (IEEE 802.1w), повторить 1-6, отметить резервные порты в п.3 и п.5,

отличие работы протокола RSTP от протокола STP в п.4